Embryo Image Preprocessing Pipeline Guide

Optical Sharpness Filtering, Centered Square Cropping, CLAHE Contrast Enhancement & Integration

1. Microscopic Imaging Challenges in IVF

Raw blastocyst microscopy presents optical artifacts that degrade standard neural network representations if unhandled:

• **Multi-Focal Z-Stack Optical Blur:** Microscopists take 2–4 slices at varying focal depths; off-axis planes obscure Inner Cell Mass boundaries.

• **Peripheral Dark Vignetting & Dish Borders:** Circular petri dishes cause severe edge shading and circular borders that confuse convolutional feature detectors.

• **Low Cellular Contrast:** Living embryonic cells have subtle refractive gradients that require local contrast amplification.

2. The 4-Stage Preprocessing Pipeline

Stage	Algorithm / Formula	Clinical Impact & Implementation
Stage 1: Sharpness Assessment	Variance of Laplacian: *Var(Laplacian(·))*	Quantifies high-frequency edge content. Automatically ranks Z-slices and picks the sharpest equatorial slice per patient.
Stage 2: Blastocyst Centering	Aspect-Preserved Square Crop: *sz = min(h, w)*	Cuts away peripheral dark microscope borders while preventing squishing or distortion of the blastocyst geometry.
Stage 3: Anti-Aliased Resizing	Lanczos-4 Interpolation to 256x256	Maintains high-order spatial frequency fidelity for delicate trophectoderm cellular junction borders.
Stage 4: Contrast Enhancement	LAB Color CLAHE: *clipLimit=2.0, tileGrid=(8,8)*	Selectively enhances luminance (L*) channel while preserving biological color integrity, making the ICM pop distinctly.

Figure 6: Step-by-Step Clinical Preprocessing Pipeline for Blastocyst Images



Figure 6: Visual 4-Stage Preprocessing Progression: (1) Raw Microscope Image → (2) Centered Square Crop → (3) Lanczos Resizing → (4) CLAHE Contrast Enhancement highlighting ICM compaction.

3. Production Code Integration for Web Developers

Import the companion module preprocessing_pipeline.py directly into your website backend handler:

```
from preprocessing_pipeline import preprocess_embryo_image, compute_sharpness, select_best_slice

# Scenario A: User uploads a single blastocyst image
clean_image = preprocess_embryo_image('user_upload.png', target_size=256)
# clean_image is ready for display on website or input to inference!

# Scenario B: User uploads multiple focal slices (Z-stack)
user_slices = ['slice_depth_1.png', 'slice_depth_2.png', 'slice_depth_3.png']
best_slice_path = select_best_slice(user_slices)
best_image = preprocess_embryo_image(best_slice_path)
```

4. Backend Deliverables Included in this Subfolder

• preprocessing_pipeline.py: Standalone library with sharpness calculation, square cropping, and CLAHE.
• inference_pipeline.py: Full end-to-end predictor that loads the 4 models and returns ready-to-display JSON for your web app.
• models/: Directory containing all 4 trained model checkpoints (best_convnext_tiny.pth, best_resnet50_full.pth, best_densenet121.pth, best_proper_efficientnet_b0.pth).
• result_images/: All 7 publication-quality PNG figures ready to display on the project website.